



2025

E23(B)

BOOKLET NO.

201106

Agriculture
Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

P.T.O.

**SECTION – A**

Q1. Write the answer of following in about **150** words. **(10×5=50)**

Q1. (a) Define cell and cell organelles. Give a list of various organelles found in plant cell. Describe mitochondria. **10**

Q1. (b) How do you differentiate Classical genetics, Molecular genetics and Population genetics ? Explain why Mendel got success. **10**

Q1. (c) How would you prove that DNA replicates in semi-conservative manner ? **10**

Q1. (d) Write significance of seed certification in seed production. **10**

Q1. (e) Differentiate between polygenic and oligogenic traits and its significance in plant breeding. **10**

Q2. (a) i) What do you mean by epistasis ? Enlist the various types of epistasis along with their ratios. Explain incomplete dominance theory. **(10×2=20)**

ii) Define chromosome. State the chromosome theory of inheritance and describe morphology on the basis of size, shape and number.

Q2. (b) i) List out the various breeding procedures in maize crop improvement and explain synthetic variety. **(10×2=20)**

ii) How to develop new male sterile lines in paddy through three line breeding approach ?

Q2. (c) Define genetic code. Write salient features of genetic code. **10**



- Q3. (a) i) Write significance of meiosis cell division. Assuming $2n = 4$ draw various stages of meiosis by diagrammatic presentation. (10×2=20)
- ii) Explain how the cultivated hexaploid wheat is evolved.
- Q3. (b) i) Elaborate the risk of transgenic technology in agriculture. (10×2=20)
- ii) Write role of Protection of Plant Varieties and Farmer's Rights Act (PPV and FR Act), 2001 in Indian agriculture. Explain how do the farmer's variety are protected.
- Q3. (c) Tabulate the difference between C3 and C4 plants. What is the biological significance of C4 cycle ? 10
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- Q4. (a) Define hybrid vigour. Write features and types of heterosis. Give in detail the reasons to get hybrid vigour. 20
- Q4. (b) List out the naturally occurring Plant Growth Regulators (PGR's) and synthetic Plant Growth Regulators (PGR's) along with their action. 20
- Q4. (c) Write different genetical and physical purity tests carried under seed certification program. Write steps in the seed certification process. 10



SECTION – B

Q5. Write the answer of following in about 150 words. (10×5=50)

Q5. (a) Explain the problems in modern vegetable farming. 10

Q5. (b) Define diagnosis and describe steps involved in plant disease diagnosis. 10

Q5. (c) Discuss the various government initiatives to ensure food security in India. 10

Q5. (d) Define Integrated Pest Management. Describe the basic principles, objectives and need of Integrated Pest Management (IFM). 10

Q5. (e) Explain the importance of protected cultivation in ornamental horticulture. 10

Q6. (a) Explain the causes for unfruitfulness in fruit crops. 20

Q6. (b) Classify insect pests on their occurrence, intensity and losses caused. Describe the factors responsible for insect pest outbreak. 20

Q6. (c) Discuss the challenges faced by Public Distribution System (PDS) in India. What measures to be adopted to made it effective and transparent ? 10

Q7. (a) Why pesticides are formulated ? Describe the types of formulations with suitable examples. 20

Q7. (b) Enumerate the factors contributing to poverty in India. What measures can be taken to reduce poverty in India ? 20

Q7. (c) Explain the processing of turmeric. 10

Q8. (a) What is the status and consequences of malnutrition in India ? How can nutrient deficiency in India be addressed ? 20

Q8. (b) Explain the necessity of adopting post harvest management practices for fruits and vegetables. 20

Q8. (c) Enlist stored grain pests. Discuss their preventive management strategy. 10
